

Kernel ridge regression

(a). $\nabla_{\theta} \|\theta\|^2$

$$\text{Let } \theta = [\theta_1, \dots, \theta_n]^T$$

$$\|\theta\|^2 = \sum_{i=1}^n \theta_i^2$$

$$\frac{\partial \|\theta\|^2}{\partial \theta_j} = 2\theta_j$$

$$\text{We have } \nabla_{\theta} \|\theta\|^2 = 2\theta$$

$$\Rightarrow \nabla_{\theta} T(\theta) = \nabla_{\theta} \left[\sum_{i=1}^n (\theta^T x^{(i)} - y^{(i)})^2 + \lambda \|\theta\|^2 \right]$$

$$= X^T X \theta - X^T \vec{y} + \lambda \theta = 0$$

$$\Rightarrow \theta = (X^T X + \lambda I)^{-1} X^T \vec{y}$$

(b). Let Φ be the design matrix associated with the feature vectors $\phi(x^{(i)})$.

According to the result of (a):

$$\theta = (\Phi^T \Phi + \lambda I)^{-1} \Phi^T \vec{y}$$

$$= \Phi^T (\Phi \Phi^T + \lambda I)^{-1} \vec{y}$$

$$= \Phi^T (K + \lambda I)^{-1} \vec{y}$$

(where K is the kernel matrix for the training set)

To predict a new value y_{new} , we can compute

$$\vec{y}_{\text{new}} = \theta^T \phi(x_{\text{new}})$$

$$= \vec{y}^T (K + \lambda I)^{-1} \Phi \phi(x_{\text{new}})$$

$$= \sum_{i=1}^n \alpha_i \underbrace{K(x^{(i)}, x_{\text{new}})}$$

$$\left(\text{where } \alpha = \vec{y}^T (K + \lambda I)^{-1} \right)$$

— all these terms can be efficiently computed using the kernel function

2. l_2 norm

(a). Consider a potential solution to the above problem with some $\xi < 0$

Then the constraint $y^{(i)}(w^T x^{(i)} + b) \geq 1 - \xi_i$ would also be satisfied for $\xi_i = 0$,

and the objective function would be lower, proving that this could not be an optimal solution.

$$(b). L(w, b, \xi, \alpha) = \frac{1}{2} w^T w + \frac{c}{2} \sum_{i=1}^m \xi_i^2 - \sum_{i=1}^m \alpha_i [y^{(i)}(w^T x^{(i)} + b) - 1 + \xi_i],$$

where $\alpha_i \geq 0$ for $i = 1, \dots, m$.

$$(c). \nabla_w L = w - \sum_{i=1}^m \alpha_i y^{(i)} x^{(i)} = 0 \Rightarrow w = \sum_{i=1}^m \alpha_i y^{(i)} x^{(i)}$$

$$\frac{\partial L}{\partial b} = - \sum_{i=1}^m \alpha_i y^{(i)} = 0 \Rightarrow \sum_{i=1}^m \alpha_i y^{(i)} = 0$$

$$\nabla_{\xi} L = c \xi - \alpha = 0 \Rightarrow c \xi_i = \alpha_i \text{ for } i = 1, \dots, m$$

(d) Based on (c)'s result, substitute w, b with α

$$\text{Max}_{\alpha} \sum_{i=1}^m \alpha_i - \frac{1}{2} \sum_{i=1}^m \sum_{j=1}^m \alpha_i \alpha_j y^{(i)} y^{(j)} (x^{(i)})^T x^{(j)} - \frac{1}{2} \sum_{i=1}^m \frac{\alpha_i^2}{c}$$

$$\text{s.t. } \alpha_i \geq 0 \quad i = 1, \dots, m$$

$$\sum_{i=1}^m \alpha_i y^{(i)} = 0$$

3. (a). We first set $\alpha_i = 1$ for all $i = 1, \dots, m$ and $b = 0$

Then, for a training example $\{x^{(i)}, y^{(i)}\}$, we get.

$$\begin{aligned}
 |f(x^{(i)}) - y^{(i)}| &= \left| \sum_{d=1}^m y^{(d)} k(x^{(d)}, x^{(i)}) - y^{(i)} \right| \\
 &= \left| \sum_{d=1}^m y^{(d)} \exp(-\|x^{(d)} - x^{(i)}\|^2 / T^2) - y^{(i)} \right| \\
 &= \left| y^{(i)} + \sum_{d \neq i} y^{(d)} \exp(\|x^{(d)} - x^{(i)}\|^2 / T^2) - y^{(i)} \right| \\
 &= \left| \sum_{d \neq i} y^{(d)} \exp(-\|x^{(d)} - x^{(i)}\|^2 / T^2) \right| \\
 &\leq \sum_{d \neq i} |y^{(d)} \exp(-\|x^{(d)} - x^{(i)}\|^2 / T^2)| \\
 &= \sum_{d \neq i} |y^{(d)}| \cdot \exp(-\|x^{(d)} - x^{(i)}\|^2 / T^2) \\
 &= \sum_{d \neq i} \exp(-\|x^{(d)} - x^{(i)}\|^2 / T^2) \\
 &\leq \sum_{d \neq i} \exp(-\xi^2 / T^2) = (m-1) \exp(-\xi^2 / T^2)
 \end{aligned}$$

Namely, $|f(x^{(i)}) - y^{(i)}| \leq (m-1) \exp(-\xi^2 / T^2)$

We want $|f(x^{(i)}) - y^{(i)}| < 1$ for each:

$$\Rightarrow (m-1) \exp(-\xi^2 / T^2) < 1$$

$$\Rightarrow T < \frac{\xi}{\log(m-1)}$$

for example, $T = \frac{\xi}{\log m}$ we are done

(b). The classifier will obtain zero training error.

The SVM without slack variables will always return zero training error if it is able to find a solution, so all that remains to be shown is that there exists one feasible point.

Consider the constraint $y^{(i)}(w^T x^{(i)} + b) \geq 1$ for some i , and let $b = 0$

$$\text{Then } y^{(i)}(w^T x^{(i)} + b) = y^{(i)} f(x^{(i)}) \geq 0$$

as guaranteed in (a)

Therefore, as we choose all the α_i 's large enough, $y^{(i)}(w^T x^{(i)} + b) \geq 1$, so the optimization problem is feasible.
 (Note $w = \sum_{i=1}^m \alpha_i y^{(i)} x^{(i)}$).

(c). Not necessarily.

$$\frac{1}{2} \|w\|^2 + C \sum_{i=1}^m \xi_i$$

The C parameter controls the relative weight of the two terms of the SVM training objective.

If the C parameter is sufficiently small, then the latter component will have relatively little contribution to the objective.

In this case, a weight vector which has a very small norm but does not achieve zero training error may achieve a lower objective value than one which achieves zero training error.

5 Uniform Convergence Note: in this scenario $\min_i \epsilon(h_i) = 0$

(a). Let $h \in H$ be a hypothesis with true error greater than γ

Then $P("h \text{ predicts correctly"}) \leq 1 - \gamma$

$$= P("h \text{ predicts correctly } m \text{ times}") \leq (1 - \gamma)^m \leq e^{-\gamma m}$$

Applying the union bound.

$$P(\exists h \in H, \text{ s.t. } \epsilon(h) > \gamma \wedge h \text{ predicts correctly } m \text{ times}) \leq k e^{-\gamma m}$$

We want to make this probability equal to δ .

$$k e^{-\gamma m} = \delta$$

$$\Rightarrow \gamma = \frac{1}{m} \log \frac{k}{\delta}$$

This implies that with probability $1 - \delta$,

$$\epsilon(\hat{h}) \leq \frac{1}{m} \log \frac{k}{\delta}$$

$$(b) \quad k e^{-\gamma m} = \delta \Rightarrow m = \frac{1}{\gamma} \log \frac{k}{\delta}$$